Mid-Term Exam-Spring 2026-AEM 428/528

We are intercepting an incoming missile launched from 50 km away using a defensive missile. The mass, launch angle, and nozzle exit Mach number of the defensive missile are treated as design variables. Our objective is to choose these variables so that the interception time is minimized.

The interception time to minimize is the time it takes for the defense missile to reach to the specified point and meet with the attack missile. So, we wait and launch the defense missile after the attack missile has been launched. The time for the defense missile to intercept the attack missile is the interception time.

The incoming missile is to be intercepted at a point located 1 km horizontally from the target. The interception altitude will be determined from the trajectory calculations of the incoming missile. Thus, the interception point is defined by x=49 km and y equal to the altitude of the incoming missile when it reaches x= 49 km.

The specifications of the incoming and defensive missiles are given below.

Grading:

You will get 32 points if you just attempt the problem. There are 6 plots and 10 variables to be determined in the problem.

Scoring:

- Each complete plot earns you 5 pts. Each plot with missing figure details earns you 3 points. Total of 30 points.
- Each variable of the part a (attack missile part) is 3 points -- total of 18 points
- Each variable of part b (defense missile part) is 5 points. Total of 20 points.

30+18+20 = 68

What to turn in:

- The figures must be no smaller than half a page.
- Both the abscissa and ordinate must be clearly labeled, including the proper units.
- Use Times Roman font, size 14, for all numbers and text.
- Include both major and minor grid lines.
- Make sure the axis limits begin and end at logical values, and that the plotted data occupies a substantial portion of the figure. For example, do not set an axis from 9.23 to 20.01 unless there is a clear reason to do so.
- The first page must include a brief problem statement, along with drawings or pictures, and the values of the requested variables.
- You must also submit the MATLAB code used for the work.
- Upload both the MATLAB code and the exam-related document to Blackboard.

Attack missile specifications:

Mi = 1670;	initial mass [kg]
Mf = 570;	final mass (dry mass) [kg]
Mp = Mi - Mf;	propellant mass [kg]
Cd = 0.45;	drag coefficient
Dref = 0.61;	diameter of the missile to use in drag area calculations [m]
P0 = 101325*20;	chamber stagnation pressure [Pa]
T0 = 2500;	chamber stagnation temperature [K]
Dexit= 0.58;	exit diameter of the CD nozzle [m]
gamma= 1.24;	ratio of specific heats
Rgas = 350;	nozzle gas constant [J/(kg-K)]
Mexit= 3;	exit Mach number

Assumptions:

- The missile is launched from the ground level to target the 50 km away from the launch location.
- You will need to consider the variation of the pressure, density and temperature with elevation. A simple code is given at the bottom of the document.
- Earth is flat, there is no wind, g=9.80665 m/s^2
- Use dt increments of 0.0001s
- The missile is not guided; thus, its trajectory is a gravity turn trajectory. In the gravity turn launch the thrust and the velocity vectors are in the same direction, on the same line of action.

Calculate:

a) the throat area of the CD nozzle
b) the mass flow rate of the propellant if the P0, and T0 values are constants
c) the exit temperature
d) the exit velocity
e) the launch angle with respect to the horizontal axis for the missile to land at exactly 50 km distance
f) time of flight till impact (if the missile was not intercepted)

Plot (abscissa is time):

1. the variation of the missile path angle in time
2. the variation of the missile elevation in time
3. the variation of the missile x component velocity in time
4. the variation of the missile y component velocity in time
5. the variation of the missile acceleration in time
6. the variation of the missile thrust in time

Defense missile specifications:

Mi=70	initial mass [kg]
Mf = 35 – 60,	final mass (dry mass) [kg]
Mp = Mi - Mf;	propellant mass [kg]
Cd = 0.45;	drag coefficient
Dref = 0.15;	diameter of the missile to use in drag area calculations [m]
P0 = 101325*20;	chamber stagnation pressure [Pa]
T0 = 2500;	chamber stagnation temperature [K]
Dexit= 0.13;	exit diameter of the CD nozzle [m]
gamma= 1.24;	ratio of specific heats
Rgas = 350;	nozzle gas constant [J/(kg-K)]
Mexit= 2-3;	exit Mach number

You want to intercept the attack missile with a lighter missile. The defense missile is launched from the target location, and it is also a gravity turn missile. Intercept point is the location where the attack missile at x = 49 km. The y location is determined from the trajectory of the attack missile.

- You will need to choose the final mass, launch angle and exit Mach number to minimize the intercept time. The initial missile mass is given as 70 kg. The ranges to use for the mass and the exit Mach number is listed in the table above. You are free to choose the launch angle. You want to intercept the missile within 0.5 m of the intercept point determined.

Calculate:

a. Final mass
b. Launch angle
c. Exit Mach number

d. Time of flight to intercept

Additional information:

```
---- atmosphere ----
Tsl  = 288.15;          % [K]
Psl  = 101325;          % [Pa]
L    = 0.0065;          % [K/m]
Rair = 287.0;           % [J/(kg-K)]

if h(i) <= 11000
   Tatm = Tsl - L*h(i);
   Patm = Psl * (Tatm/Tsl)^(g0/(Rair*L));
else
   Tatm = 216.65;
   P11  = Psl * (Tatm/Tsl)^(g0/(Rair*L));
   Patm = P11 * exp(-g0*(h(i)-11000)/(Rair*Tatm));
end
rho = Patm/(Rair*Tatm);
```

ATACMS MISSILE

NV NEW VOICE

Platforms:

M270



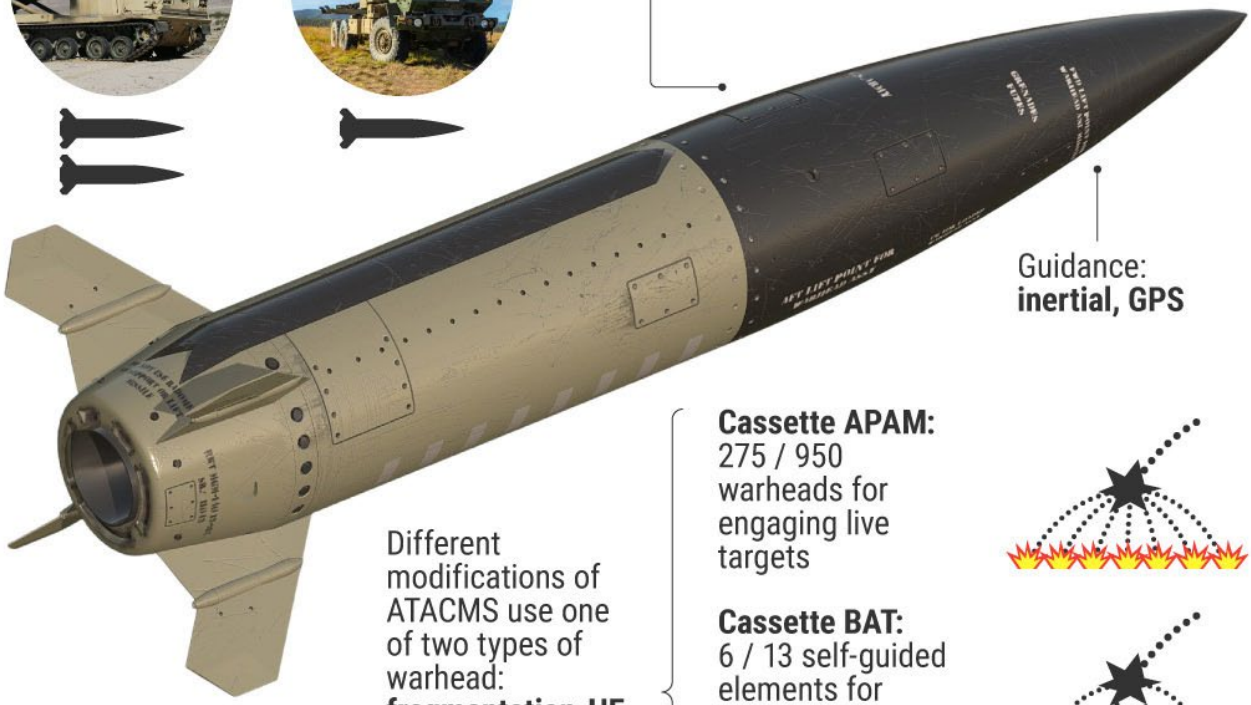
HIMARS



LOCKHEED MARTIN

Cost per unit – **~$1 million**

USA



Guidance: **inertial, GPS**

Different modifications of ATACMS use one of two types of warhead: **fragmentation-HE or cassette**

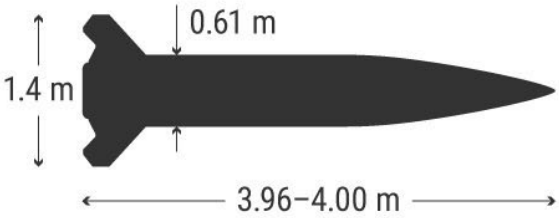
Cassette APAM: 275 / 950 warheads for engaging live targets

Cassette BAT: 6 / 13 self-guided elements for engaging armored targets



Standard warhead: WDU18 – weighing 227 kg





In service	Weight	Warhead weight	Range	Practical range	Speed	Circular deviation
since 1991	1.67 tons	160–560 kg	140–300 km	50 km	Mach 3	up to 10 m